

Homological Approach to Hybrid Dynamics II

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Abstract

I will describe results and future work towards the long term goal of characterizing the dynamics of hybrid/discontinuous systems from data.

As I will explain a fundamental challenge is that in general the conjugacy class of a smooth dynamical system cannot be learned. In contrast, I will briefly describe a theoretical result that proves that given sufficient data and sufficient computational resources it is possible to characterize global dynamics using a combinatorial/topological framework.

In practice, the cost of carrying out such computations grows exponentially with dimension. To address this I will discuss current work that makes use of latent dynamics to understand high dimensional systems. In particular, I will state a theoretical result and show a few examples that indicate the combinatorial/topological approach applied to latent dynamics works much better than expected.

Both of the above mentioned efforts are based on the underlying dynamics being continuous. Lack of continuity is one of the potential features of hybrid dynamics. With this in mind we apply our techniques to a simple model of a vibro-impact energy harvester which has explicit nonlinearities. I will discuss our results and a novel approach for identifying discontinuities.

There is very little theoretical justification for the success of our method when applied to the vibro-impact energy harvester. Thus, we plan to develop a mathematical understanding of capabilities of the combinatorial/topological framework in the context of discontinuous systems.

Time permitting I will discuss ongoing work to develop a purely algebraic framework for nonlinear dynamics.

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